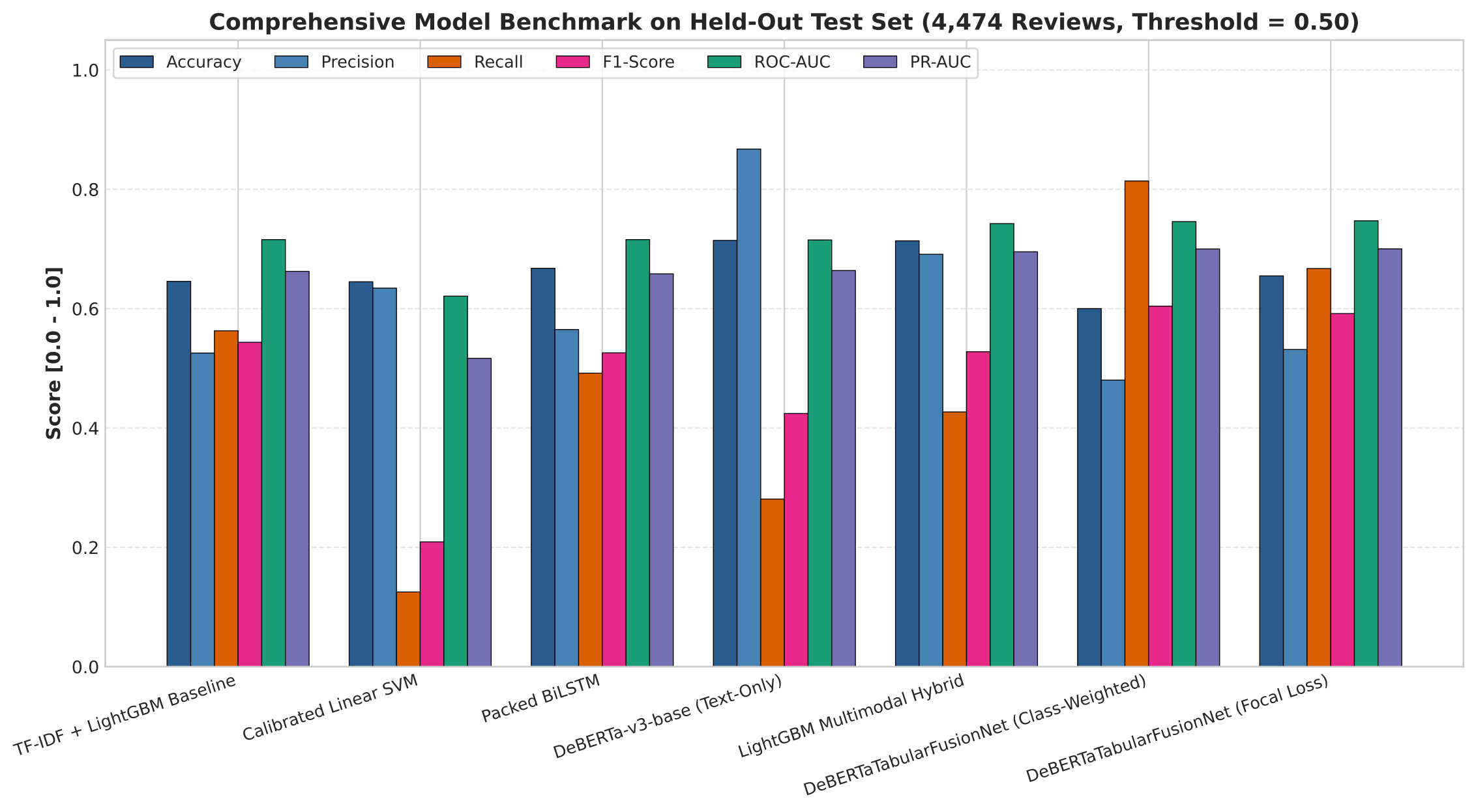
Comprehensive Model Benchmark on Held-Out Test Set (4,474 Reviews, Threshold = 0.50)
Accuracy
Precision
Recall
F1-Score
ROC-AUC
PR-AUC
Score [0.0 - 1.0]
1.0
0.8
0.6
0.4
0.2
0.0
TF-IDF + LightGBM Baseline
Calibrated Linear SVM
Packed BiLSTM
DeBERTa-v3-base (Text-Only)
LightGBM Multimodal Hybrid
DeBERTaTabularFusionNet (Class-Weighted)
DeBERTaTabularFusionNet (Focal Loss)